

Gaussian Integral

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1 Gaussian Integral

Theorem 1.1.

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$$

Proof

Defining I

$$I = \int_{-\infty}^{\infty} e^{-ax^2} dx$$

Squaring I

$$I^2 = \int_{-\infty}^{\infty} e^{-ax^2} dx \cdot \int_{-\infty}^{\infty} e^{-ay^2} dy$$

Combining the integrals

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-ax^2-ay^2} dx dy$$

Converting to Polar coordinates

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-ar^2} r dr d\theta$$

Solving

$$I^2 = \int_0^{2\pi} \left. \frac{-1}{2a} e^{-ar^2} \right|_0^{\infty} d\theta$$

$$I^2 = \int_0^{2\pi} \frac{1}{2a} d\theta$$

$$I^2 = 2\pi \frac{1}{2a}$$

$$I^2 = \frac{\pi}{a}$$

$$I = \sqrt{\frac{\pi}{a}}$$

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$$

Since e^{-ax^2} is an even function we also get the integral

$$\int_0^{\infty} e^{-ax^2} dx = \frac{\sqrt{\frac{\pi}{a}}}{2} = \sqrt{\frac{\pi}{4a}} \quad (1)$$

2 Gaussian Type Integrals

Using $\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$ we can take derivatives with respect to (w.r.t.) a

$$\begin{aligned}\frac{\partial}{\partial a} \int_{-\infty}^{\infty} e^{-ax^2} dx &= \frac{\partial}{\partial a} \sqrt{\frac{\pi}{a}} \\ \int_{-\infty}^{\infty} -x^2 e^{-ax^2} dx &= \sqrt{\pi} \frac{-1}{2} a^{-3/2} \\ \boxed{\int_{-\infty}^{\infty} x^2 e^{-ax^2} dx} &= \boxed{\frac{1}{2a} \sqrt{\frac{\pi}{a}}}\end{aligned}$$

This process can be repeated to obtain the general identity

$$\boxed{\int_{-\infty}^{\infty} x^{2n} e^{-ax^2} dx} = \boxed{\frac{\pi}{|\Gamma(\frac{1}{2} - n)| a^n \sqrt{a}}} \quad (2)$$